

```
In [1]: #dictionary
mydict = dict() # empty dictionary
mydict
```

```
Out[1]: {}
```

```
In [2]: mydict={}
mydict
```

```
Out[2]: {}
```

```
In [3]: mydict = {1:'one' , 2:'two' , 3:'three'} # dictionary with integer keys
mydict
```

```
Out[3]: {1: 'one', 2: 'two', 3: 'three'}
```

```
In [5]: mydict=dict({1:'one' , 2:'two' , 3:'three'})
mydict
```

```
Out[5]: {1: 'one', 2: 'two', 3: 'three'}
```

```
In [6]: mydict = {'A':'one' , 'B':'two' , 'C':'three'} # dictionary with character keys
mydict
```

```
Out[6]: {'A': 'one', 'B': 'two', 'C': 'three'}
```

```
In [7]: mydict = {1:'one' , 'A':'two' , 3:'three'} # dictionary with mixed keys
mydict
```

```
Out[7]: {1: 'one', 'A': 'two', 3: 'three'}
```

```
In [8]: mydict.keys()
```

```
Out[8]: dict_keys([1, 'A', 3])
```

```
In [9]: mydict.values()
```

```
Out[9]: dict_values(['one', 'two', 'three'])
```

```
In [10]: mydict.items()
```

```
Out[10]: dict_items([(1, 'one'), ('A', 'two'), (3, 'three')])
```

```
In [12]: mydict={1:'one',2:'two','A':['asif','john','maria']}
mydict
```

```
Out[12]: {1: 'one', 2: 'two', 'A': ['asif', 'john', 'maria']}
```

```
In [13]: mydict = {1:'one' , 2:'two' , 'A':{'Name':'asif' , 'Age' :20}, 'B':('Bat' , 'cat')}
mydict
```

```
Out[13]: {1: 'one', 2: 'two', 'A': {'Name': 'asif', 'Age': 20}, 'B': ('Bat', 'cat')}
```

```
In [15]: keys={'a','b','c','d'}
mydict3=dict.fromkeys(keys)
```

```
mydict3
```

```
Out[15]: {'a': None, 'c': None, 'd': None, 'b': None}
```

```
In [21]: keys={'a','b','c','d'}  
value=[10,20,30]  
mydict3=dict.fromkeys(keys,value)  
mydict3
```

```
Out[21]: {'a': [10, 20, 30], 'c': [10, 20, 30], 'd': [10, 20, 30], 'b': [10, 20, 30]}
```

```
In [23]: value.append(40)  
value
```

```
Out[23]: [10, 20, 30, 40, 40]
```

```
In [24]: mydict3
```

```
Out[24]: {'a': [10, 20, 30, 40, 40],  
          'c': [10, 20, 30, 40, 40],  
          'd': [10, 20, 30, 40, 40],  
          'b': [10, 20, 30, 40, 40]}
```

```
In [25]: #accessing items  
mydict = {1:'one' , 2:'two' , 3:'three' , 4:'four'}  
mydict
```

```
Out[25]: {1: 'one', 2: 'two', 3: 'three', 4: 'four'}
```

```
In [26]: mydict[1]    # Access item using key
```

```
Out[26]: 'one'
```

```
In [27]: # Access item using get() method  
mydict.get(1)
```

```
Out[27]: 'one'
```

```
In [28]: mydict1 = {'Name':'Asif' , 'ID': 74123 , 'DOB': 1991 , 'job' : 'Analyst'}  
mydict1
```

```
Out[28]: {'Name': 'Asif', 'ID': 74123, 'DOB': 1991, 'job': 'Analyst'}
```

```
In [32]: mydict1['Name']
```

```
Out[32]: 'Asif'
```

```
In [40]: mydict.get('job')
```

```
In [42]: mydict1 = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Address' : 'Hilsinki'}  
mydict1
```

```
Out[42]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki'}
```

```
In [44]: mydict1['DOB']=1992  
mydict1['Adress']='Delhi'  
mydict1
```

```
Out[44]: {'Name': 'Asif',  
         'ID': 12345,  
         'DOB': 1992,  
         'Address': 'Hilsinki',  
         'Adress': 'Delhi'}
```

```
In [46]: dict={'DOB':1995}  
mydict1.update(dict)  
mydict1
```

```
Out[46]: {'Name': 'Asif',  
         'ID': 12345,  
         'DOB': 1995,  
         'Address': 'Hilsinki',  
         'Adress': 'Delhi'}
```

```
In [47]: mydict1['job']='analyst'  
mydict1
```

```
Out[47]: {'Name': 'Asif',  
         'ID': 12345,  
         'DOB': 1995,  
         'Address': 'Hilsinki',  
         'Adress': 'Delhi',  
         'job': 'analyst'}
```

```
In [48]: mydict1.pop('job')
```

```
Out[48]: 'analyst'
```

```
In [53]: mydict1
```

```
Out[53]: {'Name': 'Asif', 'DOB': 1995, 'Address': 'Hilsinki'}
```

```
In [50]: mydict1.popitem()
```

```
Out[50]: ('Adress', 'Delhi')
```

```
In [55]: del[mydict1['Name']]  
mydict1
```

```
Out[55]: {'DOB': 1995, 'Address': 'Hilsinki'}
```

```
In [56]: #copy dictionary  
mydict = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Address' : 'Hilsinki'}  
mydict
```

```
Out[56]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki'}
```

```
In [57]: mydict1=mydict
```

```
In [58]: id(mydict),id(mydict1)
```

```
Out[58]: (1544766126528, 1544766126528)
```

```
In [59]: mydict2 = mydict.copy() # Create a copy of the dictionary
```

```
In [60]: id(mydict2)
```

```
Out[60]: 1544765664896
```

```
In [61]: mydict['Address']='Mumnai'
```

```
In [62]: mydict
```

```
Out[62]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Mumnai'}
```

```
In [64]: #Loop
mydict1 = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Address' : 'Hilsinki' }
mydict1
```

```
Out[64]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki'}
```

```
In [66]: for i in mydict:
          print(i,mydict[i])
```

```
Name Asif
ID 12345
DOB 1991
Address Mumnai
```

```
for i in mydict: print(mydict[i])
```

```
In [67]: #dictionary membership
mydict1 = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Job': 'Analyst'}
mydict1
```

```
Out[67]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Job': 'Analyst'}
```

```
In [68]: 'Name' in mydict
```

```
Out[68]: True
```

```
In [69]: 'Asif' in mydict
```

```
Out[69]: False
```

```
In [70]: # Membership test can be only done for keys.
```

```
In [ ]: All / Any
The all() method returns:
True - If all all keys of the dictionary are true
False - If any key of the dictionary is false
The any() function returns True if any key of the dictionary is True. If not, an
```

```
In [71]: mydict1 = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Job': 'Analyst'}
mydict1
```

```
Out[71]: {'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Job': 'Analyst'}
```

```
In [72]: all(mydict)
```

```
Out[72]: True
```

```
In [73]: any(mydict)
```

```
Out[73]: True
```

```
In [74]: all('Name')
```

```
Out[74]: True
```

```
In [75]: all('Sataus')
```

```
Out[75]: True
```

```
In [76]: any('Name')
```

```
Out[76]: True
```

```
In [77]: any('Status')
```

```
Out[77]: True
```